Exercise J-M

1. If the system of equations
$x + 2y + 3z = 3$
$4x + 3y - 4z = 4$
$8x + 4y - \lambda z = 9 + \mu$
has infinitely many solutions, then the ordered pair (λ, μ) is equal to : **[JEE-Main,Jan.2023]**

यदि समीकरणों
$x + 2y + 3z = 3$
$4x + 3y - 4z = 4$
$8x + 4y - \lambda z = 9 + \mu$
के अनंत हल है, तब क्रमित युग्म (λ, μ) बराबर है : **[JEE-Main,Jan.2023]**

(A) $\left(\frac{72}{5}, \frac{21}{5}\right)$ (B) $\left(-\frac{72}{5}, -\frac{21}{5}\right)$ (C) $\left(\frac{72}{5}, -\frac{21}{5}\right)$ (D) $\left(-\frac{72}{5}, \frac{21}{5}\right)$

Ans. C

Sol. $x + 2y + 3z = 3$ _____(i)
$4x + 3y - 4z = 4$ _____(ii)
$8x + 4y - \lambda z = 9 + \mu$ _____(iii)
(i) $\times 4$ – (ii) $\Rightarrow 5y + 16z = 8$ _____(iv)
(ii) $\times 2$ – (iii) $\Rightarrow 2y + (\lambda - 8)z = -1 - \mu$ _____(v)
(iv) $\times 2$ – (iii) $\times 5 \Rightarrow (32 - 5(\lambda - 8))z = 16 - 5(-1 - \mu)$

For infinite solutions $\Rightarrow 72 - 5\lambda = 0 \Rightarrow \lambda = \frac{72}{5}$

$21 + 5\mu = 0 \Rightarrow \mu = \frac{-21}{5}$

$\Rightarrow (\lambda, \mu) \equiv \left(\frac{72}{5}, \frac{-21}{5}\right)$

2. Let S_1 and S_2 be respectively the sets of all $a \in R - \{0\}$ for which the system of linear equations :
$ax + 2ay - 3az = 1$
$(2a + 1)x + (2a + 3)y + (a + 1)z = 2$
$(3a + 5)x + (a + 5)y + (a + 2)z = 3$
has unique solution and infinitely many solutions. Then : **[JEE-Main,Jan.2023]**

(A) S_1 is an infinite set and $n(S_2) = 2$ (B) $S_1 = R - \{0\}$ and $S_2 = \phi$

(C) $n(S_1) = 2$ and S_2 is an infinite set (D) $S_1 = \phi$ and $S_2 = R - \{0\}$

माना सभी $a \in R - \{0\}$, जिनके लिए रैखिक समीकरण निकाय
$ax + 2ay - 3az = 1$
$(2a + 1)x + (2a + 3)y + (a + 1)z = 2$
$(3a + 5)x + (a + 5)y + (a + 2)z = 3$

का केवल एक हल है तथा अनंत हल है, के समुच्चय क्रमशः S_1 तथा S_2 है। तो **[JEE-Main,Jan.2023]**

(A) S_1 एक अपरिमित समुच्चय है तथा $n(S_2) = 2$ है

(B) $S_1 = R - \{0\}$ तथा $S_2 = \phi$

(C) $n(S_1) = 2$ तथा S_2 एक अपरिमित समुच्चय है

(D) $S_1 = \phi$ तथा $S_2 = R - \{0\}$

Ans. B

Sol. $$D = \begin{bmatrix} a & 2a & -3a \\ 2a+1 & 2a+3 & 2a+1 \\ 3a+5 & a+5 & a+2 \end{bmatrix} = a\left(15a^2 + 31a + 37\right)$$

$\Rightarrow D \neq a$ for all $a \in R - \{0\}$

3. Consider the following system of questions

$$\alpha x + 2y + z = 1$$
$$2\alpha x + 3y + z = 1$$
$$3x + \alpha y + 2z = \beta$$

for some $\alpha, \beta \in R$. Then which of the following is NOT correct. **[JEE-Main,Jan.2023]**

(A) It has no solution for $\alpha = -1$ and for all $\beta \in R$

(B) It has no solution for $\alpha = 3$ and for all $\beta \neq 2$

(C) It has no solution if $\alpha = -1$ and $\beta \neq 2$

(D) It has no solution for all $\alpha \neq -1$ and $\beta = 2$

किसी $\alpha, \beta \in R$ के लिए निम्न समीकरण निकाय का विचार कीजिए :

$$\alpha x + 2y + z = 1$$
$$2\alpha x + 3y + z = 1$$
$$3x + \alpha y + 2z = \beta$$

तब निम्न में से कौन सा सही नहीं है। **[JEE-Main,Jan.2023]**

(A) $\alpha = -1$ एवं सभी $\beta \in R$ के लिए इसका कोई हल नहीं है

(B) $\alpha = 3$ एवं सभी $\beta \neq 2$ के लिए इसका कोई हल नहीं है

(C) इसका कोई हल नहीं है यदि $\alpha = -1$ एवं $\beta \neq 2$ हैं

(D) सभी $\alpha \neq -1$ एवं $\beta = 2$ के लिए इसका एक हल है

Ans. A

Sol. $D=\begin{vmatrix}\alpha & 2 & 1\\ 2\alpha & 3 & 1\\ 3 & \alpha & 2\end{vmatrix}=0\Rightarrow\alpha=-1,3$

$$D_x=\begin{vmatrix}2 & 1 & 1\\ 3 & 1 & 1\\ \alpha & 2 & \beta\end{vmatrix}=0\Rightarrow\beta=2$$

$$D_y=\begin{vmatrix}\alpha & 1 & 1\\ 2\alpha & 1 & 1\\ 3 & 2 & \beta\end{vmatrix}=0$$

$$D_z=\begin{vmatrix}\alpha & 2 & 1\\ 2\alpha & 3 & 1\\ 3 & \alpha & \beta\end{vmatrix}=0$$

$\beta = 2,\ \alpha = -1$
$\alpha = -1,\ \beta = 2$Infinite solution

4. Let the system of linear equations

$x + y + kz = 2$

$2x + 3y - z = 1$

$3x + 4y + 2z = k$

have infinitely many solutions. Then the system

$(k + 1)x + (2k - 1)y = 7$

$(2k + 1)x + (k + 5)y = 10$

has : **[JEE-Main,Jan.2023]**

(A) no solution

(B) unique slution satisfying $x - y = 1$

(C) infinitely many solutions

(D) unique solution satisfying $x + y = 1$

माना रैखिक समीकरण निकाय

$x + y + kz = 2$

$2x + 3y - z = 1$

$3x + 4y + 2z = k$

के अनंत हल है। तो निकाय

$(k + 1)x + (2k - 1)y = 7$

$(2k + 1)\,x + (k + 5)\,y = 10$ **[JEE-Main,Jan.2023]**

(A) का कोई हल नहीं हैं

(B) केवल एक हल है जो $x - y = 1$ को संतुष्ट करता है

(C) के अनंत हल है

(D) केवल एक हल है जो $x + y = 1$ को संतुष्ट करता है

Ans. D

Sol. $\begin{vmatrix} 1 & 1 & k \\ 2 & 3 & -1 \\ 3 & 4 & 2 \end{vmatrix} = 0$

$\Rightarrow 1(10) - 1(7) + k(-1) - 0$

$\Rightarrow k = 3$

For $k = 3$, 2nd system is

$$4x + 5y = 7 \quad ...(1)$$

and $$7x + 8y = 10 \quad ...(2)$$

Clearly, they have a unique solution

(B) – (1) $\Rightarrow 3x + 3y = 3$

$\Rightarrow x + y = 1$

5. For $\alpha, \beta \in R$, suppose system of linear equations

$x - y + z = 5$

$2a + 2y + \alpha z = 8$

$3x - y + 4z = \beta$

has infinitely many solutions. Then α and β are the roots of: **[JEE-Main,Jan.2023]**

$\alpha, \beta \in R$ के लिए, माना समीकरण निकाय

$x - y + z = 5$

$2a + 2y + \alpha z = 8$

$3x - y + 4z = \beta$

के अनंत हल है। तब α व β निम्न में से किसके मूल है : **[JEE-Main,Jan.2023]**

(A) $x^2 - 10x + 16 = 0$ (B) $x^2 + 18x + 56 = 0$

(C) $x^2 + 14x + 24 = 0$ (D) $x^2 - 18x + 56 = 0$

Ans. D

Sol. $\begin{vmatrix} 1 & -1 & 1 \\ 2 & 2 & \alpha \\ 3 & -1 & 4 \end{vmatrix} = 0$; $8 + \alpha - 2(-4 + 1) + 3(-\alpha - 2) = 0$

$8 + \alpha + 6 - 3\alpha - 6 = 0$

$\alpha = 4$

6. For the system of linear equations

$x + y + z = 6$

$\alpha x + \beta y + 7z = 3$

$x + 2y + 3z = 14,$

which of the following is **NOT** true? **[JEE-Main,Jan.2023]**

(A) If $\alpha = \beta = 7$, then the system has no solution.

(B) If $\alpha = \beta$ and $\alpha \neq 7$, then the system has a unique solution.

(C) There is a unique point (α, β) on the line $x + 2y + 18 = 0$ for which the system has infinitely many solutions.

(D) For every point $(\alpha, \beta) \neq (7, 7)$ on the line $x - 2y + 7 = 0$, the system has infinitely many solutions.

रैखिक समीकरण निकाय

$x + y + z = 6$

$\alpha x + \beta y + 7z = 3$

$x + 2y + 3z = 14,$

के लिए निम्न में से कौन सा सत्य नहीं है ? **[JEE-Main,Jan.2023]**

(A) यदि $\alpha = \beta = 7$ है, तो निकाय का कोई हल नहीं है

(B) यदि $\alpha = \beta$ तथा $\alpha \neq 7$ है, तो निकाय का केवल एक हल है

(C) रेखा $x + 2y + 18 = 0$ पर केवल एक बिंदु (α, β) है जिसके लिए निकाय के अनंत हल है

(D) रेखा $x - 2y + 7 = 0$ पर प्रत्येक बिंदु $(\alpha, \beta) \neq (7, 7)$ के लिए निकाय के अनंत हल है

Ans. D

Sol. By equation 1 and 3 $y + 2z = 8$

$y = 8 - 2z$

And $x = -2 + z$

Now putting in equation 2

$\alpha(z - 2) + \beta(-2z + 8) + 7z = 3$

$\Rightarrow (\alpha - 2\beta + 7)z = 2\alpha - 8\beta + 3$

So equations have unique solution if

$\alpha - 2\beta + 7 \neq 0$

And equations have no solution if

$\alpha - 2\beta + 7 = 0$ and $2\alpha - 8\beta + 3 \neq 0$

And equations have infinite solution if

$\alpha - 2\beta + 7 = 0$ and $2\alpha - 8\beta + 3 = 0$

7. Let S denote the set of all real values of λ such that the system of equations

$\lambda x + y + z = 1$
$x + \lambda y + z = 1$
$x + y + \lambda z = 1$

is inconsistent, then $\sum_{\lambda \in S}\left(|\lambda|^2 + |\lambda|\right)$ is equal to : **[JEE-Main,Jan.2023]**

माना λ के सभी वास्तविक मानों, जिसके लिए समीकरण निकाय

$\lambda x + y + z = 1$
$x + \lambda y + z = 1$
$x + y + \lambda z = 1$

असंगत है, का समुच्चय S है, तब $\sum_{\lambda \in S}\left(|\lambda|^2 + |\lambda|\right)$ का मान है : **[JEE-Main,Jan.2023]**

(A) 4 (B) 6 (C) 2 (D) 12

Ans. B

Sol. $D = \begin{vmatrix} \lambda & 1 & 1 \\ 1 & \lambda & 1 \\ 1 & 1 & \lambda \end{vmatrix} = (\lambda - 1)^2(\lambda + 2)$ If $D = 0 \Rightarrow \lambda = 1, -2$

$D_1 = \begin{vmatrix} 1 & 1 & 1 \\ 1 & \lambda & 1 \\ 1 & 1 & \lambda \end{vmatrix} = (\lambda - 1)^2$ When $\lambda = -2$, $D \neq 0$

when $\lambda = 1$, all equations are identical so number of solutions are infinite.
So for inconsistent system, λ can be equal to -2 only.

$\therefore \sum_{\lambda \to 5}\left(|\lambda|^2 + |\lambda|\right) = |-2|^2 + |-2| = 4 + 2 = 6$

8. Let $f(x) = \begin{vmatrix} 1+\sin^2 x & \cos^2 x & \sin 2x \\ \sin^2 x & 1+\cos^2 x & \sin 2x \\ \sin^2 x & \cos^2 x & 1+\sin 2x \end{vmatrix}$, $x \in \left[\frac{\pi}{6}, \frac{\pi}{3}\right]$. If α and β respectively are the maximum and the minimum values of f, then : **[JEE-Main,Jan.2023]**

माना $f(x)=\begin{vmatrix}1+\sin^2 x & \cos^2 x & \sin 2x\\ \sin^2 x & 1+\cos^2 x & \sin 2x\\ \sin^2 x & \cos^2 x & 1+\sin 2x\end{vmatrix}$, $x\in\left[\frac{\pi}{6},\frac{\pi}{3}\right]$ हैं यदि f के अधिकतम व न्यूनतम मान क्रमशः α व β हैं, तब :

(A) $\beta^2+2\sqrt{\alpha}=\frac{19}{4}$ (B) $\alpha^2-\beta^2=4\sqrt{3}$ (C) $\beta^2-2\sqrt{\alpha}=\frac{19}{4}$ (D) $\alpha^2+\beta^2=\frac{19}{2}$ **[JEE-Main,Jan.2023]**

Ans. C

Sol. $f(x)=\begin{vmatrix}1+\sin^2 x & \cos^2 x & \sin 2x\\ \sin^2 x & 1+\cos^2 x & \sin(2x)\\ \sin^2 x & \cos^2 x & 1+\sin(2x)\end{vmatrix}, x\in\left[\frac{\pi}{6},\frac{\pi}{3}\right]$

$R_1\to R_1-R_2$
$R_2\to R_2-R_3$

$$f(x)=\begin{vmatrix}1 & -1 & 0\\ 0 & 1 & -1\\ \sin^2 x & \cos^2 x & 1+\sin(2x)\end{vmatrix}$$

$= 1\ (1+\sin(2x))+\cos^2x+\sin^2x$

$f(x)=2+\sin 2x$ $\quad \begin{bmatrix} x\in\left(\frac{\pi}{6},\frac{\pi}{3}\right)\\ 2x\in\left(\frac{\pi}{3},\frac{2\pi}{3}\right)\end{bmatrix}$

$\max(f(x))=3=\alpha$

$\text{Min}\,(f(x))=2+\frac{\sqrt{3}}{2}=\beta$

$\therefore \beta^2=\left(2+\frac{\sqrt{3}}{2}\right)^2=4+\frac{3}{4}+2\sqrt{3}=\frac{19}{4}+2\sqrt{3}\quad \therefore \beta^2-2\sqrt{\alpha}=\frac{19}{4}+2\sqrt{3}-2\sqrt{3}=\frac{19}{4}$

9. For the system of linear equations $\alpha x+y+z=1$, $x+\alpha y+z=1$, $x+y+\alpha z=\beta$, which one of the following statements is NOT true? **[JEE-Main,Jan.2023]**

(A) It has infinitely many solutions if $\alpha=1$ and $\beta=1$

(2) $x+y+z=\frac{3}{4}$ if $\alpha=2$ and $\beta=1$

(3) It has no solution if $\alpha=-2$ and $\beta=1$

(4) It has infinitely many solutions if $\alpha=2$ and $\beta=-1$

रैखिक समीकरण निकाय $\alpha x + y + z = 1,\ x + \alpha y + z = 1,\ x + y + \alpha z = \beta$ के लिए निम्न में से कौन सा कथन सही नहीं है ?

[JEE-Main,Jan.2023]

(A) इसके अनंत हल है यदि $\alpha = 1$ तथा $\beta = 1$ है

(2) $x + y + z = \frac{3}{4}$ यदि $\alpha = 2$ तथा $\beta = 1$ हैं

(3) इसका कोई हल नहीं है यदि $\alpha = -2$ तथा $\beta = 1$ हैं

(4) इसके अनंत हल है यदि $\alpha = 2$ तथा $\beta = -1$ हैं

Ans. D

Sol. $\begin{vmatrix} \alpha & 1 & 1 \\ 1 & \alpha & 1 \\ 1 & 1 & \alpha \end{vmatrix} = \alpha(\alpha^2 - 1) - (\alpha - 1) + (1 - \alpha) = \alpha^3 - 3\alpha + 2 = (\alpha - 1)(\alpha^2 + \alpha - 2)$

$(\alpha - 1)(\alpha - 1)(\alpha + 2)$ θ.f $\alpha = 2, \beta = -1$ then

for $\alpha = 1$ system of equations is $\theta x + y + z = 1$

$x + y + z = 1$ $x + 2y + z = 1$

$x + y + z = 1$ $x + y + 2z = -1$ No solution

$x + y + z = \beta \Rightarrow \beta = 1$ then infinitely many solutions

10. If the system of equations

$x + y + az = b$

$2x + 5y + 2z = 6$

$x + 2y + 3z = 3$

has infinitely many solutions, then $2a + 3b$ is equal to : **[JEE-Main,April,2023]**

यदि समीकरण निकाय

$x + y + az = b$

$2x + 5y + 2z = 6$

$x + 2y + 3z = 3$

के अनंत हल है। तब $2a + 3b$ बराबर है। **[JEE-Main,April,2023]**

(A) 20 (B) 25 (C) 28 (D) 23

Ans. D

Sol. $x + y + az = b$

$2x + 5y + 2z = 6$

$x + 2y + 3z = 3$

For ∞ solution

$\Delta = 0, \Delta_x = 0, \Delta_y = 0, \Delta_z = 0$

$\Delta = \begin{vmatrix} 1 & 1 & a \\ 2 & 5 & 2 \\ 1 & 2 & 3 \end{vmatrix} = 0 \Rightarrow 11 - 4 - a = 0 \Rightarrow a = 7$

$$\Delta_z = \begin{vmatrix} 1 & 1 & b \\ 2 & 5 & 6 \\ 1 & 2 & 3 \end{vmatrix} = 0 \Rightarrow 3 - 0 - b = 0 \Rightarrow b = 3$$

Hence $2a + 3b = 23$

11. For the system of equations

$x + y + z = 6$

$x + 2y + \alpha z = 10$

$x + 3y + 5z = \beta$, which one of the following is NOT true? **[JEE-Main,April,2023]**

(A) System has a unique solution for $\alpha = -3, \beta = 14$

(B) System has a unique solution for $\alpha = -3, \beta \neq 14$

(C) System has no solution for $\alpha = 3, \beta = 24$

(D) System has infinitely many solutions for $\alpha = 3, \beta = 14$

समीकरण निकाय

$x + y + z = 6$

$x + 2y + \alpha z = 10$

$x + 3y + 5z = \beta$, के लिए निम्न में से कौन सा सत्य नहीं है ? **[JEE-Main,April,2023]**

(A) $\alpha = -3, \beta = 14$ के लिए समीकरण निकाय का अद्वितीय हल नहीं

(B) $\alpha = -3, \beta \neq 14$ के लिए समीकरण निकाय का अद्वितीय हल है

(C) $\alpha = 3, \beta = 24$ के लिए समीकरण निकाय का कोई हल नहीं है

(D) $\alpha = 3, \beta = 14$ के लिए समीकरण निकाय के अनंत हल है

Ans. D

Sol. $$D = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & \alpha \\ 1 & 3 & 5 \end{vmatrix}$$

$= 1(10 - 3\alpha) - (5 - \alpha) + (3 - 2)$

$= 6 - 2\alpha$

$D \neq 0 \Rightarrow \alpha \neq 3$

Unique solution $\Rightarrow \alpha \neq 3$

12. Let S be the set of all values of $\theta \in [-\pi, \pi]$ for which the system of linear equations

$x + y + \sqrt{3}z = 0$

$-x + (\tan\theta)y + \sqrt{7}z = 0$

$x + y + (\tan\theta)z = 0$

has non-trivial solution. Then $\frac{120}{\pi}\sum_{\theta \in S}\theta$ is equal to : **[JEE-Main,April,2023]**

माना $\theta \in [-\pi, \pi]$ के सभी मानों, जिनके लिए रैखिक समीकरण निकाय

$x + y + \sqrt{3}z = 0$

$-x + (\tan\theta)y + \sqrt{7}z = 0$

$x + y + (\tan\theta)z = 0$

का अतुच्छ हल है, का समुच्च S है। तो $\frac{120}{\pi}\sum_{\theta\in S}\theta$ बराबर है : **[JEE-Main,April,2023]**

(A) 30 (B) 20 (C) 10 (D) 40

Ans. B

Sol. For non trivial solutions

$D = 0$

$$\begin{vmatrix} 1 & 1 & \sqrt{3} \\ -1 & \tan\theta & \sqrt{7} \\ 1 & 1 & \tan\theta \end{vmatrix} = 0$$

$\tan^2\theta - \left(\sqrt{3}-1\right)\tan\theta - \sqrt{3} = 0$

$\tan\theta = \sqrt{3}, -1$

$\theta = \left\{\frac{\pi}{3}, \frac{-2\pi}{3}, \frac{-\pi}{4}, \frac{3\pi}{4}\right\}$

$\frac{120}{\pi}\left(\sum\theta\right) = \frac{120}{\pi}\times\frac{\pi}{6} = 20$

13. For the system of linear equations

$2x - y + 3z = 5$
$3x + 2y - z = 7$
$4x + 5y + \alpha z = \beta,$

which of the following is NOT correct? **[JEE-Main,April,2023]**

(A) The system has infinitely many solutions for $\alpha = -5$ and $\beta = 9$

(B) The system has infinitely many solutions for $\alpha = -6$ and $\beta = 9$

(C) The system has a unique solution for $\alpha \neq -5$ and $\beta = 8$

(D) The system is inconsistent for $\alpha = -5$ and $\beta = 8$

रैखिक समीकरण निकाय

$2x - y + 3z = 5$
$3x + 2y - z = 7$
$4x + 5y + \alpha z = \beta,$

के लिए निम्न में से कौन सा सही नहीं है ? **[JEE-Main,April,2023]**

(A) $\alpha = -5$ तथा $\beta = 9$ के लिए समीकरण निकाय के अनंत हल हैं

(B) $\alpha = -6$ तथा $\beta = 9$ के लिए समीकरण निकाय के अनंत हल हैं

(C) $\alpha \neq -5$ तथा $\beta = 8$ के लिए समीकरण निकाय का एक अद्वितीय हल है

(D) $\alpha = -5$ तथा $\beta = 8$ के लिए समीकरण निकाय असंगत है

Ans. B

Sol. For infinite solution

$\lambda_1 P_1 + \lambda_2 P_2 = P_3$

$\lambda_1(2x - y + 3z - 5) + (3x + 2y - z - 7)\lambda_2 = (4x + 5y + \alpha z - \beta)$

$2\lambda_1 + 3\lambda_2 = 4] \quad \lambda_1 = -1$

$-\lambda_1 + 2\lambda_2 = 5] \quad \lambda_2 = 2$

$3\lambda_1 - \lambda_2 = \alpha \rightarrow \alpha = -5$

$-5\lambda_1 - 7\lambda_2 = -\beta \rightarrow \beta = 9$

14. If $\begin{vmatrix} x+1 & x & x \\ x & x+\lambda & x \\ x & x & x+\lambda^2 \end{vmatrix} = \frac{9}{8}(103x+81)$, then $\lambda, \frac{\lambda}{3}$ are the roots of the equation :

[JEE-Main,April,2023]

यदि $\begin{vmatrix} x+1 & x & x \\ x & x+\lambda & x \\ x & x & x+\lambda^2 \end{vmatrix} = \frac{9}{8}(103x+81)$ है, तो $\lambda, \frac{\lambda}{3}$ किस समीकरण के मूल है : **[JEE-Main,April,2023]**

(A) $4x^2 - 24x + 27 = 0$ (B) $4x^2 - 24x - 27 = 0$

(C) $4x^2 + 24x + 27 = 0$ (D) $4x^2 + 24x - 27 = 0$

Ans. A

Sol. $\begin{vmatrix} x+1 & x & x \\ x & x+\lambda & x \\ x & x & x+\lambda^2 \end{vmatrix} = \frac{9}{8}(103x+81)$

Put $x = 0$

$\begin{vmatrix} 1 & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda^2 \end{vmatrix} = \frac{9}{8} \times 81$

$\lambda^3 = \frac{3^6}{2^3}$

$\lambda = \frac{9}{2}$

$$\frac{\lambda}{3}=\frac{3}{2}$$

Equation : $x^2-\left(\frac{9}{2}+\frac{3}{2}\right)x+\frac{9}{2}\times\frac{3}{2}=0$

$4x^2-24x+27=0$

15. If the system of linear equations
$7x + 11y + \alpha z = 13$
$5x + 4y + 7z = \beta$
$175x + 194y + 57z = 361$
has infinitely many solutions, then $\alpha + \beta + 2$ is equal to : **[JEE-Main,April,2023]**

यदि रैखिक समीकरण निकाय
$7x + 11y + \alpha z = 13$
$5x + 4y + 7z = \beta$
$175x + 194y + 57z = 361$
के अनंत हल हैं तो $\alpha + \beta + 2$ बराबर है : **[JEE-Main,April,2023]**

(A) 5 (B) 6 (C) 3 (D) 4

Ans. D

Sol. $7x+11y+\alpha z=13$ ....(i)

$5x+4y+7z=\beta$ ....(ii)

$175x+194y+57z=361$ ....(iii)

$(i)\times10+(ii)\times21-(iii)$

$z(10\alpha+147-57)=130+21\beta-361$

$\therefore 10\alpha+90=0$

$\alpha=-9$

$130-361+21\beta=0$

$\beta=11$

$\alpha+\beta+2=4$

16. For the system of linear equations
$2x + 4y + 2az = b$
$x + 2y + 3z = 4$
$2x - 5y + 2z = 8$
which of the following is NOT correct? **[JEE-Main,April,2023]**

(A) has unique solution if $a = b = 8$

(B) It has infinitely many solutions if $a = 3, b = 8$

(C) It has infinitely many solutions if $a = 3, b = 6$

(D) It has unique solution if $a = b = 6$

रैखिक समीकरण निकाय

$2x + 4y + 2az = b$

$x + 2y + 3z = 4$

$2x - 5y + 2z = 8$

के लिए निम्न में से कौनसा सही नहीं है? **[JEE-Main,April,2023]**

(A) इसका अद्वितीय हल है यदि $a = b = 8$ हैं

(B) इसके अनंत हल हैं यदि $a = 3, b = 8$ हैं

(C) इसके अनंत हल हैं यदि $a = 3, b = 6$ हैं

(D) इसका अद्वितीय हल है यदि $a = b = 6$ हैं

Ans. C

Sol. $$D = \begin{vmatrix} 2 & 4 & 2a \\ 1 & 2 & 3 \\ 2 & -5 & 2 \end{vmatrix}$$

$= 2(4 + 15) - 4(2 - 6) + 2a(-5 - 4)$

$= 38 + 16 - 18a = 54 - 18a = 18(3 - a)$

$$D_1 = \begin{vmatrix} b & 4 & 2a \\ 4 & 2 & 3 \\ 8 & -5 & 2 \end{vmatrix}$$

$= b(4 + 15) - 4(8 - 24) + 2a(-20 - 16)$

$= 19b + 64 - 72a$

$= 19b - 72a + 64$

$$D_2 = \begin{vmatrix} 2 & b & 2a \\ 1 & 4 & 3 \\ 2 & 8 & 2 \end{vmatrix}$$

$= 2(8 - 24) - b(2 - 6) + 2a(8 - 8)$

$= -32 + 4b = 4(b - 8)$

$$D_3 = \begin{vmatrix} 2 & 4 & b \\ 1 & 2 & 4 \\ 2 & -5 & 8 \end{vmatrix}$$

$= 2(16 + 20) - 4(8 - 8) + b(-5 - 4)$

$= 72 - 9b = 9(8 - b)$

Now when $a \neq 3, b \in R$ system have unique solution.

when $a = 3, b \neq 8$, system have no solution

when $a = 3, b = 8$, system have infinitely many solutions.

17. If the system of equations

$2x + y - z = 5$

$2x - 5y + \lambda z = \mu$

$x + 2y - 5z = 7$

has infinitely many solutions, then $(\lambda+\mu)^2 + (\lambda-\mu)^2$ is equal to : **[JEE-Main,April,2023]**

यदि समीकरण निकाय

$2x + y - z = 5$
$2x - 5y + \lambda z = \mu$
$x + 2y - 5z = 7$

के अनंत हल हैं, तो $(\lambda+\mu)^2 + (\lambda-\mu)^2$ बराबर है : **[JEE-Main,April,2023]**

(A) 916 (B) 904 (C) 912 (D) 920

Ans. A

Sol. $2x + y - z = 5$

$2x - 5y + \lambda z = \mu$

$x + 2y - 5z = 7$

for infinite solution $D = D_1 = D_2 = D_3 = 0$

$$D = \begin{vmatrix} 2 & 1 & -1 \\ 2 & -5 & \lambda \\ 1 & 2 & -5 \end{vmatrix} = 0$$

$2(25 - 2\lambda) - (-10 - \lambda) - 1(4 + 5) = 0$

$50 - 4\lambda + 10 + \lambda - 9 = 0 \quad \Rightarrow \lambda = 17$

$$D_1 = \begin{vmatrix} 5 & 1 & -1 \\ \mu & -5 & 17 \\ 7 & 2 & -5 \end{vmatrix} = 0$$

$5(25 - 34) - \mu(-5 + 2) + 7(17 - 5) = 0$

$-45 + 3\mu + 84 = 0 \Rightarrow \mu = -13$

Now $(\lambda+\mu)^2 + (\lambda-\mu)^2 = (17-13)^2 + (17+13)^2$

$= 16 + 900 = 916$

18. Let the system of linear equations

$-x + 2y - 9z = 7$
$-x + 3y + 7z = 9$
$-2x + y + 5z = 8$
$-3x + y + 13z = \lambda$

has a unique solution $x = \alpha$, $y = \beta$, $z = \gamma$. Then the distance of the point(α, β, γ) from the plane $2x - 2y + z = \lambda$ is : **[JEE-Main,April,2023]**

यदि रेखिक समीकरण निकाय

$-x + 2y - 9z = 7$
$-x + 3y + 7z = 9$
$-2x + y + 5z = 8$
$-3x + y + 13z = \lambda$

के अद्वितीय हल $x = \alpha, y = \beta, z = \gamma$ हो, तो बिन्दु(α, β, γ) की समतल $2x - 2y + z = \lambda$ से दूरी होगी :

[JEE-Main,April,2023]

(A) 13 (B) 9 (C) 11 (D) 7

Ans. D

Sol. $-x + 2y - 9z = 7$ ...(1)

$-x + 3y + 7z = 9$ ...(2)

$-2x + y + 5z = 8$ ...(3)

(B) – (1)

$y + 16z = 2$ ...(4)

(C) $- 2 \times$ (1)

$-3y + 23z = -6$ ...(5)

$3 \times (4) + (5)$

$71z = 0 \Rightarrow z = 0$

$y = 2$

$x = -3$

$(-3, 2, 0) \rightarrow (\alpha, \beta, \gamma)$

Put in $-3x + y + 13z = \lambda$

$\lambda = 9 + 2 = 11$

$$d = \left|\frac{-6-4-11}{3}\right| = 7$$

19. Let S be the set of values of λ, for which the system of equations

$6\lambda x - 3y + 3z = 4\lambda^2$,

$2x + 6\lambda y + 4z = 1$,

$3x + 2y + 3\lambda z = \lambda$

has no solution. Then $12\sum_{\lambda \in S} |\lambda|$ is equal to _____. **[JEE-Main,April,2023]**

माना λ के मानों, जिनके लिए समीकरण निकाय

$6\lambda x - 3y + 3z = 4\lambda^2$,

$2x + 6\lambda y + 4z = 1$,

$3x + 2y + 3\lambda z = \lambda$

का कोई हल नहीं है, का समुच्चय S है। तो $12\sum_{\lambda \in S} |\lambda|$ बराबर है _____ **[JEE-Main,April,2023]**

Ans. 24

Sol. For no solution

$$\begin{vmatrix} 6\lambda & -3 & 3 \\ 2 & 6\lambda & 4 \\ 3 & 2 & 3\lambda \end{vmatrix} = 0$$

$\Rightarrow 9\lambda^3 - 7\lambda - 2 = 0$

$\Rightarrow (\lambda - 1)(3\lambda + 1)(3\lambda + 2) = 0$

$\Rightarrow \quad 12\sum_{\lambda \in S} |\lambda| = 12 \times \left(1 + \frac{1}{3} + \frac{2}{3}\right) = 24$

20. Let $D_k = \begin{vmatrix} 1 & 2k & 2k-1 \\ n & n^2+n+2 & n^2 \\ n & n^2+n & n^2+n+2 \end{vmatrix}$. If $\sum_{k=1}^{n} D_k = 96$, then n is equal to _____.

[JEE-Main,April,2023]

माना $D_k = \begin{vmatrix} 1 & 2k & 2k-1 \\ n & n^2+n+2 & n^2 \\ n & n^2+n & n^2+n+2 \end{vmatrix}$ है। यदि $\sum_{k=1}^{n} D_k = 96$ तो n बराबर है _____

[JEE-Main,April,2023]

Ans. 6

Sol. $D_k = \begin{vmatrix} 1 & 2k & 2k-1 \\ n & n^2+n+2 & n^2 \\ n & n^2+n & n^2+n+2 \end{vmatrix}$

$\sum_{k=1}^{n} D_k = 96 \Rightarrow$

$$\begin{vmatrix} \sum_{k=1}^{n} 1 & \sum 2k & \sum (2k-1) \\ n & n^2+n+2 & n^2 \\ n & n^2+n & n^2+n+2 \end{vmatrix} = 96$$

$$\Rightarrow \begin{vmatrix} n & n^2+n & n^2 \\ n & n^2+n+2 & n^2 \\ n & n^2+n & n^2+n+2 \end{vmatrix} = 96$$

$R_2 \to R_2 - R_1$ and $R_3 \to R_3 - R_1$

$$\begin{vmatrix} n & n^2+n & n^2 \\ 0 & 2 & 0 \\ 0 & 0 & n+2 \end{vmatrix} = 96$$

$\Rightarrow n(2n+4) = 96 \Rightarrow n(n+2) = 48 \Rightarrow n = 6$

21. The values of α, for which $\begin{vmatrix} 1 & \frac{3}{2} & \alpha+\frac{3}{2} \\ 1 & \frac{1}{3} & \alpha+\frac{1}{3} \\ 2\alpha+3 & 3\alpha+1 & 0 \end{vmatrix} = 0$ lie in the interval : **[JEE-Main,Jan.2024]**

α के वह मान, जिनके लिए $\begin{vmatrix} 1 & \frac{3}{2} & \alpha+\frac{3}{2} \\ 1 & \frac{1}{3} & \alpha+\frac{1}{3} \\ 2\alpha+3 & 3\alpha+1 & 0 \end{vmatrix} = 0$ है, किस अंतराल में है ? **[JEE-Main,Jan.2024]**

(A) (0, 3) (B) $\left(-\frac{3}{2},\frac{3}{2}\right)$ (C) (–2, 1) (D) (–3, 0)

Ans. D

Sol. $\begin{vmatrix} 1 & \frac{3}{2} & \alpha+\frac{3}{2} \\ 1 & \frac{1}{3} & \alpha+\frac{1}{3} \\ 2\alpha+3 & 3\alpha+1 & 0 \end{vmatrix} = 0$

$$\Rightarrow (2\alpha+3)\left\{\frac{7\alpha}{6}\right\}-(3\alpha+1)\left\{\frac{-7}{6}\right\}=0$$

$$\Rightarrow (2\alpha+3)\cdot\frac{7\alpha}{6}+(3\alpha+1)\cdot\frac{7}{6}=0$$

$$\Rightarrow 2\alpha^2+3\alpha+3\alpha+1=0$$

$$\Rightarrow 2\alpha^2+6\alpha+1=0$$

$$\Rightarrow \alpha=\frac{-3+\sqrt{7}}{2},\frac{-3-\sqrt{7}}{2}$$

22. Consider the system of linear equations $x+y+z=4\mu, x+2y+2\lambda z=10\mu, x+3y+4\lambda^2 z=\mu^2+15$, where $\lambda, \mu \in R$. Which one of the following statements is NOT correct ? **[JEE-Main,Jan.2024]**

(A) The system is consistent if $\lambda \neq \frac{1}{2}$

(B) The system is inconsistent if $\lambda = \frac{1}{2}$ and $\mu \neq 1$

(C) The system has infinite number of solutions if $\lambda = \frac{1}{2}$ and $\mu=15$

(D) The system has unique solution if $\lambda \neq \frac{1}{2}$ and $\mu \neq 1, 15$

रैखिक समीकरण निकाय $x + y + z = 4\mu, x + 2y + 2\lambda z = 10\mu, x + 3y + 4\lambda^2 z = \mu^2 + 15$, जहाँ $\lambda, \mu \in R$, है, का विचार कीजिए। निम्न कथनों में से कौनसा सही नहीं है ? **[JEE-Main,Jan.2024]**

(A) यदि $\lambda \neq \frac{1}{2}$ है, तो निकाय संगत है।

(B) यदि $\lambda = \frac{1}{2}$ तथा $\mu \neq 1$ है, तो निकाय असंगत है।

(C) यदि $\lambda = \frac{1}{2}$ तथा $\mu=15$ हैं, तो निकाय के अनन्त हल हैं।

(D) यदि $\lambda \neq \frac{1}{2}$ तथा $\mu \neq 1, 15$ हैं, तो निकाय का अद्वितीय हल है।

Ans. B

Sol. $x + y + z = 4\mu, x + 2y + 2\lambda z = 10\mu, x + 3y + 4\lambda^2 z = \mu^2 + 15$

$$\Delta = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 2\lambda \\ 1 & 3 & 4\lambda^2 \end{vmatrix} = (2\lambda - 1)^2$$

For unique solution $\Delta \neq 0, 2\lambda - 1 \neq 0, \left(\lambda \neq \frac{1}{2}\right)$

Let $\Delta = 0, \lambda = \frac{1}{2}$

$$\Delta_y = 0, \Delta_x = \Delta_z = \begin{vmatrix} 4\mu & 1 & 1 \\ 10\mu & 2 & 1 \\ \mu^2 + 15 & 3 & 1 \end{vmatrix}$$

$= (\mu - 15)(\mu - 1)$

For infinite solution $\lambda = \frac{1}{2}, \mu = 1$ or 15

23. Consider the system of linear equations $x + y + z = 5,\ x + 2y + \lambda^2 z = 9, x + 3y + \lambda z = \mu$, where $\lambda, \mu \in \mathbb{R}$. Then,

which of the following statement is NOT correct? **[JEE-Main,Jan.2024]**

(A) System is consistent if $\lambda \neq 1$ and $\mu = 13$

(B) System has unique solution if $\lambda \neq 1$ and $\mu \neq 13$

(C) System is inconsistent if $\lambda = 1$ and $\mu \neq 13$

(D) System has infinite number of solutions if $\lambda = 1$ and $\mu = 13$

रैखिक समीकरण निकाय $x + y + z = 5,\ x + 2y + \lambda^2 z = 9, x + 3y + \lambda z = \mu$, जहाँ $\lambda, \mu \in \mathbb{R}$ हैं, का विचार कीजिए। तो निम्न में से कौनसा कथन सत्य नहीं है ? **[JEE-Main,Jan.2024]**

(A) निकाय संगत है यदि $\lambda \neq 1$ तथा $\mu = 13$ है

(B) निकाय का अद्वितीय हल है यदि $\lambda \neq 1$ तथा $\mu \neq 13$ हैं

(C) निकाय असंगत है यदि $\lambda = 1$ तथा $\mu \neq 13$ है

(D) निकाय का अनंत हल है यदि $\lambda = 1$ तथा $\mu = 13$ हैं

Ans. B

Sol. $x + y + z = 5$ ...(1)

$x + 2y + \lambda^2 z = 9$ ...(2)

$x + 3y + \lambda z = \mu$ ...(3)

from $(1) + (3) - 2(2)$

$z\left(\lambda + 1 - 2\lambda^2\right) = (\mu - 13)$

$(\lambda + 1)(2\lambda + 1)Z = (13 + \mu)$

Infinite solution is $\lambda = 1$ & $\mu = 13$

for unique solution $\lambda \neq 1$

for number of solutions $\lambda = 1$ & $\mu \neq 13$

If $\lambda \neq 1$ & $\mu \neq 13$

In case where $\lambda = \dfrac{-1}{2}$ and $\mu \neq 13$

This will generate number of solutions.

24. If the system of linear equations

$x - 2y + z = -4$

$2x + \alpha y + 3z = 5$

$3x - y + \beta z = 3$

has infinitely many solutions, then $12\alpha + 13\beta$ is equal to : **[JEE-Main,Jan.2024]**

यदि रैखिक समीकरण निकाय

$x - 2y + z = -4$

$2x + \alpha y + 3z = 5$

$3x - y + \beta z = 3$

के अनंत हल हैं, तो $12\alpha + 13\beta$ बराबर है : **[JEE-Main,Jan.2024]**

(A) 54 (B) 64 (C) 60 (D) 58

Ans. D

Sol. $x - 2y + z = -4$

$2x + \alpha y + 3z = 5$

$3x - y + \beta z = 3$

For infinite solution $D = 0, D_1 = 0, D_2 = 0$ & $D_3 = 0$

$$D = 0 \Rightarrow \begin{vmatrix} 1 & -2 & 1 \\ 2 & \alpha & 3 \\ 3 & -1 & \beta \end{vmatrix} = 0$$

$\alpha\beta - 3\alpha + 4\beta = 17$ ...(1)

$$D_1 = \begin{vmatrix} -4 & -2 & 1 \\ 5 & \alpha & 3 \\ 3 & -1 & \beta \end{vmatrix} = 0$$

$$D_2 = \begin{vmatrix} 1 & -4 & 1 \\ 2 & 5 & 3 \\ 3 & 3 & \beta \end{vmatrix} = 0$$

$= (5\beta - 9) + 4(2\beta - 9) + 1(6 - 15) = 0$

$13\beta = 54 \Rightarrow \beta = \dfrac{54}{13}$

Put in (1)

$\dfrac{54}{13}\alpha - 3\alpha + 4 \times \dfrac{54}{13} = 17 \Rightarrow \alpha = \dfrac{1}{3}$

$12\alpha + 13\beta = 12 \times \dfrac{1}{3} + 13 \times \dfrac{54}{13} = 58$

25. If the system of equations

$2x + 3y - z = 5$

$x + \alpha y + 3z = -4$

$3x - y + \beta z = 7$

has infinitely many solutions, then $13\alpha\beta$ is equal to : **[JEE-Main,Jan.2024]**

यदि समीकरण निकाय

$2x + 3y - z = 5$

$x + \alpha y + 3z = -4$

$3x - y + \beta z = 7$

के अनंत हल हैं तो $13\alpha\beta$ बराबर है : **[JEE-Main,Jan.2024]**

(A) 1210 (B) 1220 (C) 1120 (D) 1110

Ans. C

Sol. Given system of equations,

$2x + 3y - z = 5$

$x + \alpha y + 3z = -4$

$3x - y + \beta z = 7$

For infinite solutions,

$\Delta = \Delta_1 = \Delta_2 = \Delta_3 = 0$

$$\Delta_2 = \begin{vmatrix} 2 & 5 & -1 \\ 1 & -4 & 3 \\ 3 & 7 & \beta \end{vmatrix} = 0 \Rightarrow \beta = \frac{-16}{13}$$

$$\Delta_3 = \begin{vmatrix} 2 & 3 & 5 \\ 1 & \alpha & -4 \\ 3 & -1 & 7 \end{vmatrix} = 0 \Rightarrow \alpha = -70$$

$\Rightarrow 13\alpha\beta = 1120$

26. Let the system of equations $x + 2y + 3z = 5, 2x + 3y + z = 9, 4x + 3y + \lambda z = \mu$ have infinite number of solutions. Then $\lambda + 2\mu$ is equal to : **[JEE-Main,Jan.2024]**

माना समीकरण निकाय $x + 2y + 3z = 5, 2x + 3y + z = 9, 4x + 3y + \lambda z = \mu$ के अनंत हल हैं। तो $\lambda + 2\mu$ बराबर है :

[JEE-Main,Jan.2024]

(A) 15 (B) 28 (C) 22 (D) 17

Ans. D

Sol. $x + 2y + 3z = 5$

$2x + 3y + z = 9$

$4x + 3y + \lambda z = \mu$

for infinite solution $D = D_1 = D_2 = D_3 = 0$

$$D = \begin{vmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \\ 4 & 3 & \lambda \end{vmatrix} = 0$$

$(3\lambda - 3) - 2(2\lambda - 4) + 3(6 - 12) = 0$

$\lambda = -13$

$$D_3 = \begin{vmatrix} 1 & 2 & 3 \\ 2 & 3 & 9 \\ 4 & 3 & \mu \end{vmatrix} = 0$$

$3\mu - 27 - 2(2\mu - 36) + 5(6 - 12) = 0$

$\mu = 15$

$\lambda + 2\mu = -13 + 30 = 17$

27. If the system of equations

$$x + \left(\sqrt{2}\sin\alpha\right)y + \left(\sqrt{2}\cos\alpha\right)z = 0$$

$$x + \left(\cos\alpha\right)y + \left(\sin\alpha\right)z = 0$$

$$x + \left(\sin\alpha\right)y - \left(\cos\alpha\right)z = 0$$

has a non-trivial solution, then $\alpha \in \left(0, \frac{\pi}{2}\right)$ is equal to : **[JEE-Main,April,2024]**

यदि समीकरण निकाय

$$x + \left(\sqrt{2}\sin\alpha\right)y + \left(\sqrt{2}\cos\alpha\right)z = 0$$

$$x + \left(\cos\alpha\right)y + \left(\sin\alpha\right)z = 0$$

$$x + \left(\sin\alpha\right)y - \left(\cos\alpha\right)z = 0$$

का अतुच्छ हल है, तो $\alpha \in \left(0, \frac{\pi}{2}\right)$ बराबर है : **[JEE-Main,April,2024]**

(A) $\frac{3\pi}{4}$ (B) $\frac{7\pi}{24}$ (C) $\frac{11\pi}{24}$ (D) $\frac{5\pi}{24}$

Ans. D

Sol. $$\begin{vmatrix} 1 & \sqrt{2}\sin\alpha & \sqrt{2}\cos\alpha \\ 1 & \sin\alpha & -\cos\alpha \\ 1 & \cos\alpha & \sin\alpha \end{vmatrix} = 0$$

$\Rightarrow 1-\sqrt{2}\sin\alpha\left(\sin\alpha+\cos\alpha\right)+\sqrt{2}\cos\alpha\left(\cos\alpha-\sin\alpha\right)=0$

$\Rightarrow 1+\sqrt{2}\cos 2\alpha-\sqrt{2}\sin 2\alpha=0$

$\cos 2\alpha-\sin 2\alpha=-\frac{1}{\sqrt{2}}$

$\cos\left(2\alpha+\frac{\pi}{4}\right)=-\frac{1}{2}$

$2\alpha+\frac{\pi}{4}=2n\pi\pm\frac{2\pi}{3}$

$\alpha+\frac{\pi}{8}=n\pi\pm\frac{\pi}{3}$

$n=0,$

$x=\frac{\pi}{3}-\frac{\pi}{8}=\frac{5\pi}{24}$

28. If the system of equations

$11x+y+\lambda z=-5$

$2x+3y+5z=3$

$8x-19y-39z=\mu$

has infinitely many solutions, then $\lambda^4-\mu$ is equal to : **[JEE-Main,April,2024]**

यदि समीकरण निकाय

$11x+y+\lambda z=-5$

$2x+3y+5z=3$

$8x-19y-39z=\mu$

के अनंत हल हैं, तो $\lambda^4-\mu$ बराबर है : **[JEE-Main,April,2024]**

(A) 47 (B) 49 (C) 51 (D) 45

Ans. A

Sol. $11x+y+\lambda z=-5$

$2x+3y+5z=3$

$8x-19y-39z=\mu$

for infinite sol.

$$D=\begin{vmatrix}11 & 1 & \lambda\\ 2 & 3 & 5\\ 8 & -19 & -39\end{vmatrix}=0$$

$\Rightarrow 11(-117+95)-1(-78-40)+\lambda(-38-24)$

$\Rightarrow 11(-22)+118-\lambda(62)=0$

$\Rightarrow 62\lambda=118-242$

$\Rightarrow \lambda=\frac{-124}{62}=-2$

$$D_1=\begin{vmatrix}-5 & 1 & -2\\ 3 & 3 & 5\\ \mu & -19 & -39\end{vmatrix}=0$$

$\Rightarrow -5(-117+95)-1(-117-5\mu)-2(-57-3\mu)=0$

$\Rightarrow -5(-22)+117+5\mu+114+6\mu=0$

$\Rightarrow 11\mu=-110-231=-341$

$\Rightarrow \mu=-31$

$\lambda^4-\mu=(-2)^4-(-31)=16+31=47$

29. The values of m, n, for which the system of equations
$x+y+z=4$,
$2x+5y+5z=17$,
$x+2y+mz=n$
has infinitely many solutions, satisfy the equation : **[JEE-Main,April,2024]**

m, n के मान, जिनके लिए समीकरण निकाय
$x+y+z=4$,
$2x+5y+5z=17$,
$x+2y+mz=n$
के अनंत हल हैं, इनमें से किस समीकरण को संतुष्ट करते हैं : **[JEE-Main,April,2024]**

(A) $m^2+n^2-m-n=46$ (B) $m^2+n^2-mn=39$

(C) $m^2+n^2+m+n=64$ (D) $m^2+n^2-mn=68$

Ans. B

Sol. Augmented matrix is

$$\begin{bmatrix}1 & 1 & 1 & : & 4\\ 2 & 5 & 5 & : & 17\\ 1 & 2 & m & : & n\end{bmatrix}R_2\to R_2-2R_1, R_3\to R_3-R_1$$

$$\Rightarrow \begin{bmatrix} 1 & 1 & 1 & : & 4 \\ 0 & 3 & 3 & : & 9 \\ 0 & 1 & m-1 & : & n-4 \end{bmatrix}$$

After dividing R_2 by 3 we obtain

$m - 1 = 1, n - 4 = 3$

$m = 2, n = 7$

$m^2 + n^2 - mn = 4 + 49 - 14 = 49 - 10 = 39$

30. For $\alpha, \beta \in R$ and a natural number n, let $A_r = \begin{vmatrix} r & 1 & \frac{n^2}{2}+\alpha \\ 2r & 2 & n^2-\beta \\ 3r-2 & 3 & \frac{n(3n-1)}{2} \end{vmatrix}$. Then $2A_{10} - A_8$ is :

[JEE-Main,April,2024]

$\alpha, \beta \in R$ और प्राकृतिक संख्या n के लिए, माना $A_r = \begin{vmatrix} r & 1 & \frac{n^2}{2}+\alpha \\ 2r & 2 & n^2-\beta \\ 3r-2 & 3 & \frac{n(3n-1)}{2} \end{vmatrix}$, तो $2A_{10} - A_8$ है :

[JEE-Main,April,2024]

(A) $4\alpha + 2\beta$ (B) $2n$ (C) 0 (D) $2\alpha + 4\beta$

Ans. A

Sol. $A_r = \begin{vmatrix} r & 1 & \frac{n^2}{2}+\alpha \\ 2r & 2 & n^2-\beta \\ 3r-2 & 3 & \frac{n(3n-1)}{2} \end{vmatrix}$

$R_2 \to R_2 - 2R_1$

$A_r = \begin{vmatrix} r & 1 & \frac{n^2}{2}+\alpha \\ 0 & 0 & -\beta-2\alpha \\ 3r-2 & 3 & \frac{n(3n-1)}{2} \end{vmatrix}$

$Ar = (\beta + 2\alpha)(3r - 3r + 2)$

$Ar = 4\alpha + 2\beta$

Hence $2A_{10} - A_8$

$= A_{10} = 4\alpha + 2\alpha$

31. If $\alpha \neq a, \beta \neq b, \gamma \neq c$ and $\begin{vmatrix} \alpha & b & c \\ a & \beta & c \\ a & b & \gamma \end{vmatrix} = 0$, then $\frac{a}{\alpha - a} + \frac{b}{\beta - b} + \frac{\gamma}{\gamma - c}$ is equal to :

[JEE-Main,April,2024]

यदि $\alpha \neq a, \beta \neq b, \gamma \neq c$ तथा $\begin{vmatrix} \alpha & b & c \\ a & \beta & c \\ a & b & \gamma \end{vmatrix} = 0$ है, तो $\frac{a}{\alpha - a} + \frac{b}{\beta - b} + \frac{\gamma}{\gamma - c}$ बराबर है :

[JEE-Main,April,2024]

(A) 1 (B) 0 (C) 3 (D) 2

Ans. B

Sol. $R_1 \to R_1 - R_2, R_2 \to R_2 - R_3$

$$\begin{vmatrix} \alpha - a & b - \beta & 0 \\ 0 & \beta - b & c - \gamma \\ a & b & \gamma \end{vmatrix} = 0$$

$(\alpha - a)(\gamma(\beta - b) - b(c - \gamma)) - (b - \beta)(-a(c - \gamma)) = 0$

$\gamma(\alpha - a)(\beta - b) - b(\alpha - a)(c - \gamma) + a(b - \beta)(c - \gamma)$

$$\frac{\gamma}{\gamma - c} + \frac{b}{\beta - b} + \frac{a}{\alpha - a} = 0$$

32. If the system of equations $x + 4y - z = \lambda$, $7x + 9y + \mu z = -3$, $5x + y + 2z = -1$ has infinitely many solutions, then $(2\mu + 3\lambda)$ is equal to : **[JEE-Main,April,2024]**

यदि समीकरण निकाय $x + 4y - z = \lambda$, $7x + 9y + \mu z = -3$, $5x + y + 2z = -1$ के अनंत हल हैं, तो

$(2\mu + 3\lambda)$ बराबर है : **[JEE-Main,April,2024]**

(A) 3 (B) 2 (C) –3 (D) –2

Ans. C

Sol. $x + 4y - z = \lambda$

$7x + 9y + \mu z = -3$

$5x + y + 2z = -1$

$$\begin{bmatrix} 1 & 4 & -1 \\ 7 & 9 & \mu \\ 5 & 1 & 2 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \lambda \\ -3 \\ -1 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 4 & -1 \\ 7 & 9 & \mu \\ 5 & 1 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} \lambda \\ -3 \\ -1 \end{bmatrix}$$

$AX = B$

$$X = A^{-1}B = \frac{\text{adj}A}{|A|}B$$

If $|A| = 0$ and $(\text{adj}A) \cdot B = 0$, system has infinitely many solutions.

$|A| = 18 - \mu - 4(14 - 5\mu) - 1(7 - 45) = 0$

$\Rightarrow 18 - \mu - 56 + 20\mu + 38 = 0$

$\Rightarrow 19\mu = 0$

$\Rightarrow \mu = 0$

$$\text{Also adj}A = \begin{bmatrix} 18 & -9 & 9 \\ -14 & 7 & -7 \\ -38 & 19 & -19 \end{bmatrix}$$

$(\text{adj}A) \cdot B = 0$

$$\begin{bmatrix} 18 & -9 & 9 \\ -14 & 7 & -7 \\ -38 & 19 & -19 \end{bmatrix}\begin{bmatrix} \lambda \\ -3 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$18\lambda + 27 - 9 = 0$

$\Rightarrow 18\lambda = -18$

$\Rightarrow \lambda = -1$

$\Rightarrow 2\mu + 3\lambda = 3(-1) = -3$

33. Let $\lambda, \mu \in \mathbf{R}$. If the system of equations

$3x + 5y + \lambda z = 3$

$7x + 11y - 9z = 2$

$97x + 155y - 189z = \mu$

has infinitely many solutions, then $\mu + 2\lambda$ is equal to : **[JEE-Main,April,2024]**

माना $\lambda, \mu \in \mathbf{R}$ है। यदि समीकरण निकाय

$3x + 5y + \lambda z = 3$

$7x + 11y - 9z = 2$
$97x + 155y - 189z = \mu$

के अनंत हल हैं, तो $\mu + 2\lambda$ बराबर है : **[JEE-Main,April,2024]**

(A) 27 (B) 24 (C) 25 (D) 22

Ans. C

Sol. $3x + 5y + \lambda z = 3$
$7x + 11y - 9z = 2$
$97x + 155y - 189z = \mu$

By carmer's rule, $\Delta = \Delta_1 = \Delta_2 = \Delta_3 = 0$

$$\begin{vmatrix} 3 & 5 & \lambda \\ 7 & 11 & -9 \\ 97 & 155 & -189 \end{vmatrix} = 0$$

$R_3 \to R_3 - 14R_2$

$$\begin{vmatrix} 3 & 5 & \lambda \\ 7 & 11 & -9 \\ -1 & 1 & -63 \end{vmatrix} = 0$$

$C_1 \to C_1 + C_2$

$$2\begin{vmatrix} 4 & 5 & \lambda \\ 9 & 11 & -9 \\ 0 & 1 & -63 \end{vmatrix} = 0$$

$-(-36 - 9\lambda) - 63(44 - 45) = 0$
$36 + 9\lambda + 63 = 0$
$9\lambda = -99$
$\lambda = -11$
$\Delta_3 = 0$

$$\begin{vmatrix} 3 & 5 & 3 \\ 7 & 11 & 2 \\ 97 & 155 & \mu \end{vmatrix} = 0$$

$C_2 \to C_2 + C_1$

$$\begin{vmatrix} 0 & 2 & 3 \\ 5 & 4 & 2 \\ 97-\mu & 58 & \mu \end{vmatrix} = 0$$

$C_2 \to C_2 + C_1$

$$\begin{vmatrix} 0 & 2 & 3 \\ 5 & 4 & 2 \\ 97-\mu & 58 & \mu \end{vmatrix} = 0$$

$-5(2\mu - 174) + (97 - \mu)(4 - 12) = 0$

$10\mu - 870 + 776 - 8\mu = 0$

$2\mu = 94$

$\mu = 47$

$\mu + 2\lambda = 47 - 22$

$= 25$

34. Let $\alpha\beta\gamma = 45$; $\alpha, \beta, \gamma \in R$. If $x(\alpha, 1, 2) + y(1, \beta, 2) + z(2, 3, \gamma) = (0, 0, 0)$ for some $x, y, z \in R$, $x\,y\,z \neq 0$, then $6\alpha + 4\beta + \gamma$ is equal to _____. **[JEE-Main,April,2024]**

माना $\alpha\beta\gamma = 45$; $\alpha, \beta, \gamma \in R$ है। यदि कुछ $x, y, z \in R$, $x\,y\,z \neq 0$, के लिए $x(\alpha, 1, 2) + y(1, \beta, 2) + z(2, 3, \gamma) = (0, 0, 0)$ है, तो $6\alpha + 4\beta + \gamma$ बराबर है _____

[JEE-Main,April,2024]

Ans. 55

Sol. $\alpha\beta\gamma = 45$, $\alpha\beta\gamma \in R$

$x(\alpha, 1, 2) + y(1, \beta, 2) + z(2, 3, \gamma) = (0, 0, 0)$

$x, y, z \in R$, $xyz \neq 0$

$\alpha x + y + 2z = 0$

$x + \beta y + 3z = 0$

$2x + 2y + \gamma z = 0$

$xyz \neq 0 \Rightarrow$ non-trivial

$$\begin{vmatrix} \alpha & 1 & 2 \\ 1 & \beta & 3 \\ 2 & 2 & \gamma \end{vmatrix} = 0$$

$\Rightarrow \alpha(\beta\gamma - 6) - 1(\gamma - 6) + 2(2 - 2\beta) = 0$

$\Rightarrow \alpha\beta\gamma - 6\alpha - \gamma + 6 + 4 - 4\beta = 0$

$\Rightarrow 6\alpha + 4\beta + \gamma = 55$

35. If the system of equations

$$2x + 7y + \lambda z = 3$$
$$3x + 2y + 5z = 4$$
$$x + \mu y + 32z = -1$$

has infinitely many solutions, then $(\lambda - \mu)$ is equal to _____. **[JEE-Main,April,2024]**

यदि समीकरणों का निकाय

$$2x + 7y + \lambda z = 3$$
$$3x + 2y + 5z = 4$$
$$x + \mu y + 32z = -1$$

अनन्त हल रखता है, तो $(\lambda - \mu)$ बराबर है _____ **[JEE-Main,April,2024]**

Ans. 38

Sol. $2x + 7y + \lambda z = 3$

$3x + 2y + 5z = 4$

$x + \mu y + 32z = -1$

has infinitely many solution

$D = D_x = D_y = D_z = 0$

$$D_z = \begin{vmatrix} 2 & 7 & \lambda \\ 3 & 2 & 4 \\ 1 & \mu & 1 \end{vmatrix} = 0$$

$-4 + 9\mu + 28 - 6 - 8\mu + 21 = 0$

$$D = \begin{vmatrix} 2 & 7 & \lambda \\ 3 & 2 & 4 \\ 1 & -39 & 32 \end{vmatrix} = 0$$

$\mu = -39$

$\Rightarrow \lambda = -1$

$\therefore \lambda - \mu - 1 + 39 = 38$